# EV Market Performance and Insights

**Project Report: Data Analytics with Tableau**

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## 1. Introduction

### 1.1 Project Overview

The electric vehicle (EV) industry is one of the fastest-growing sectors of the global automotive market, driven by environmental awareness, government incentives and advances in battery technology. This project analyses EV market data (sales, pricing, battery specifications, driving range, manufacturing country, customer ratings and CO2 savings) and presents it through two interactive Tableau dashboards and a data story.

- **Dashboard 1 (Market Performance):** KPI cards (total EV sales 1.55B, average price $64,921, average range 324.54 km, 738K tons CO2 saved per year), production output by country and vehicle category, top manufacturers by customer review and sales units, driving range distribution, and 3-year retained value.
- **Dashboard 2 (Technology and Impact):** sales trend over model years, battery technology performance, charging vs efficiency, and regional environmental impact with CO2 saved.

Tableau Public link:
https://public.tableau.com/views/HarshalParetadataanalyticswithtableauSmartbridge/Dashboard1?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

### 1.2 Purpose

The purpose is to turn raw EV market data into clear, interactive insights that help manufacturers, consultants and researchers understand which vehicle categories drive sales, which battery technologies dominate, how range and charging compare, and where the environmental benefit is greatest, so they can make informed strategic decisions.

## 2. Ideation Phase

### 2.1 Problem Statement

EV manufacturers and analysts generate large volumes of sales, battery, manufacturing and customer data, but without a consolidated view it is hard to monitor KPIs such as annual sales, battery capacity, driving range, customer satisfaction, manufacturing distribution and market share. Three stakeholders need tailored answers:

| Persona | Role | Need |
|---|---|---|
| Emily Johnson | EV Market Strategy Director | Top categories and manufacturers, customer satisfaction, battery adoption, range comparison and high-growth segments for portfolio planning |
| David Chen | Automotive BI Manager | Manufacturer market share, sales, battery distribution, manufacturing country analysis and ratings to guide clients entering the EV market |
| Dr. Sarah Williams | Sustainable Mobility Research Analyst | Battery comparison, range improvements, global manufacturing distribution and satisfaction to support evidence-based policy |

### 2.2 Empathy Map Canvas

| Quadrant | Emily (Strategy Director) | David (BI Manager) | Dr. Williams (Researcher) |
|---|---|---|---|
| Says | "Which segments should we invest in?" | "Who holds the competitive edge?" | "Which technology drives adoption?" |
| Thinks | Needs executive-level, fast insights | Clients want proof, not opinions | Policy needs credible evidence |
| Does | Reviews category sales and satisfaction | Benchmarks manufacturers and regions | Compares batteries, range and CO2 impact |
| Feels | Pressure to optimise the portfolio | Needs confidence in recommendations | Motivated by sustainability goals |

### 2.3 Brainstorming

Ideas considered and grouped by theme:

- **Market:** sales by category, trend over years, market share by manufacturer.
- **Product:** driving range by category, 3-year retained value, price KPIs.
- **Technology:** battery type vs sales and cycle life, charging power vs efficiency.
- **Sustainability:** CO2 saved by region and drivetrain, production by country.
- **Selected:** two dashboards (Market Performance, Technology and Impact) joined by a Tableau story.

## 3. Requirement Analysis

### 3.1 Customer Journey Map

| Stage | User action | Dashboard support |
|---|---|---|
| Awareness | Opens the dashboard to see the market at a glance | KPI cards on Dashboard 1 |
| Exploration | Filters by vehicle category and compares charts | Category colour legend and filters |
| Analysis | Drills into manufacturers, range, battery and charging | Bubble, box, bar and scatter charts |
| Decision | Picks segments, technologies and regions to focus on | Retained value, battery and CO2 views |
| Sharing | Presents findings to stakeholders | Tableau Public link and story |

### 3.2 Solution Requirement

**Functional requirements**

- Display headline KPIs: total sales, average price, average range, CO2 saved.
- Compare sales, production and retained value across vehicle categories and countries.
- Show manufacturer performance by customer review and sales units.
- Compare battery technologies by sales units and cycle life.
- Show charging power vs efficiency and regional CO2 savings.
- Provide interactive filtering by vehicle category.

**Non-functional requirements**

- Clear, readable layout with consistent colours for categories.
- Responsive interaction and fast loading.
- Publicly shareable through Tableau Public.

### 3.3 Data Flow Diagram

EV dataset (CSV/Excel) → Tableau data source connection → data cleaning and calculated fields (sales, retained value, CO2 saved) → worksheets (charts) → Dashboard 1 and Dashboard 2 → Story → Tableau Public.

### 3.4 Technology Stack

| Layer | Tool |
|---|---|
| Data source | EV market dataset (CSV/Excel) |
| Visualisation | Tableau Desktop |
| Publishing | Tableau Public |

| Layer | Tool |
|---|---|
| Documentation | Microsoft Word / Claude Docs |

## 4. Project Design

### 4.1 Problem Solution Fit

The problem is scattered EV data with no single view for decision makers. The solution is a pair of focused interactive dashboards that answer each persona's questions directly, so insights that would take hours in raw tables are available in one screen.

### 4.2 Proposed Solution

- **Dashboard 1:** market KPIs, production by country and category, manufacturer reviews vs sales, driving range distribution, retained value.
- **Dashboard 2:** sales trend, battery technology performance, charging vs efficiency, regional CO2 impact map.
- **Story:** a guided sequence of both dashboards with key observations.
- **Delivery:** published on Tableau Public for easy sharing.

### 4.3 Solution Architecture

Data layer (EV dataset) → Tableau data model with calculated fields → worksheet layer (15 category-coloured charts) → dashboard layer (Dashboard 1 and Dashboard 2) → story layer → Tableau Public for end users (Emily, David, Dr. Williams).

## 5. Project Planning & Scheduling

### 5.1 Project Planning

| Phase | Activities | Output |
|---|---|---|
| 1. Understanding | Study problem statement and personas | Business questions |
| 2. Data preparation | Load dataset, clean fields, create calculated fields | Tableau data source |
| 3. Visualisation | Build worksheets for each business question | Worksheets |
| 4. Dashboards | Combine worksheets into two dashboards | Dashboard 1 and 2 |
| 5. Story and publishing | Build story, publish to Tableau Public | Public link |
| 6. Documentation | Complete report and templates | Final report |

## 6. Functional and Performance Testing

### 6.1 Performance Testing

| Test | Expected result | Status |
|---|---|---|
| KPI cards display correct totals | Sales 1.55B, price $64,921, range 324.54 km, CO2 738K tons | Pass |
| Category colour legend filter | Charts update for selected categories | Pass |
| Tooltips on charts | Values shown on hover | Pass |
| Dashboard load on Tableau Public | Both dashboards render fully | Pass |
| Navigation between dashboards | Tabs switch without error | Pass |

## 7. Results

### 7.1 Output Screenshots

*Insert the Dashboard 1 and Dashboard 2 screenshots here.*

**Key findings**

- Total EV sales reach 1.55B units at an average price of $64,921, with an average range of 324.54 km and 738K tons of CO2 saved per year.
- India, China and Vietnam lead production output.
- Electric Bus has the highest 3-year retained value (14.5M), followed by Electric Truck, Sports Car (8.69M) and Luxury SUV (6.15M).
- Luxury Sedans and SUVs offer the longest driving range; micro city cars and two/three-wheelers the shortest.
- NMC and LFP batteries lead in sales units; Solid-State and LFP show the longest cycle life.
- Higher-capacity vehicles charge fastest (about 350 kW) but use more energy per km.
- Annual sales are stable from 2018 to 2026, led by Two-Wheeler Scooters.

## 8. Advantages & Disadvantages

**Advantages**

- Single interactive view of sales, technology, range and environmental impact.
- Serves several personas with one project.
- Easy to share and explore through Tableau Public.
- Supports data-driven product and investment decisions.

**Disadvantages**

- Depends on the quality and coverage of the dataset.
- Tableau Public data is publicly visible.
- Some charts are dense with 15 vehicle categories.

- Data is a snapshot and does not update in real time.

## 9. Conclusion

The project shows how Tableau can convert EV market data into actionable insight. The dashboards reveal which categories and manufacturers lead, which battery technologies dominate, how range and charging trade off, and where CO2 savings are greatest. Together they give strategy, consulting and research users a clear basis for planning, positioning and policy.

## 10. Future Scope

- Connect live or regularly refreshed data sources.
- Add forecasting of sales and battery adoption.
- Include charging infrastructure and government incentive data.
- Add manufacturer-level drill-downs and market share views.
- Build mobile-optimised dashboard layouts.

## 11. Appendix

**Source Code (if any):** Not applicable (Tableau workbook).

**Dataset Link:** *used local .csv dataset*

**GitHub & Project Demo Link:** Demo: https://public.tableau.com/views/HarshalParetadataanalyticswithtableauSmartbridge/Dashboard1?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

Github: https://github.com/HarshalPareta148/dataAnalyticsWithTableau